

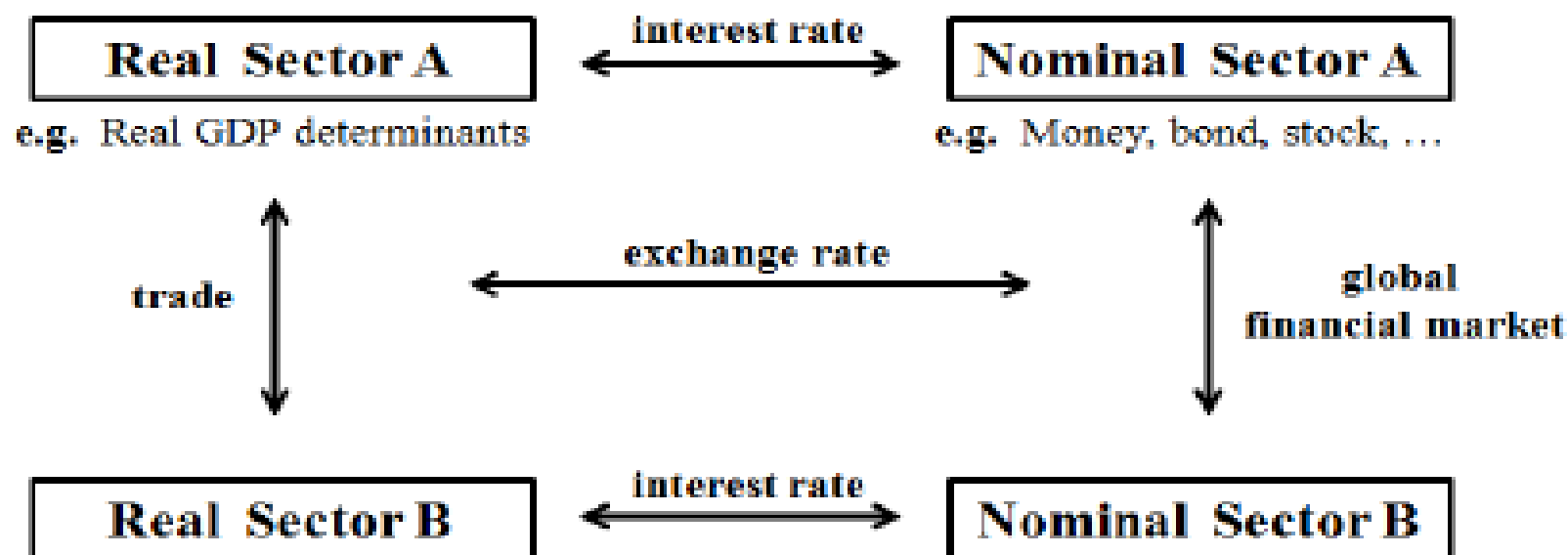
국제 금융 기초

기본 모델

□ 2국가 모델

- Two countries, A and B
- Each economy comprises of nominal and real sectors
- Strong interactions among the sectors across countries

Figure 1.1: Two Sectors of Economy



기본 모델

□ 2국가 모델

- Key questions

- Why exchange rates matter?
- How to forecast exchange rates?

Arbitrage-free Exchange Rate

Arbitrage-free Exchange Rate

- ☐ Partial Equilibrium Approach
- ☐ Exchange rate: price 1 USD in terms of KRW

Absolute PPP

□ 절대 구매력 평가설

- Assumptions

1. Two countries (KOR, US)
2. All goods are identical and tradable
3. No transaction cost
4. CPI weights are the same across the countries

- Notations

- P_i = unit price of good i in terms of KRW
- P_i^* = unit price of good i in terms of USD
- S = unit price of USD in terms of KRW (*exchange rate*)
- W_i = CPI weight of good i
- P = CPI of KOR
- P^* = CPI of US

Absolute PPP

□ 절대 구매력 평가설

• Derivation

- By assumptions 1, 2, and 3, we have

$$P_i = P_i^* \times S$$

for $\forall_i = 1, 2, \dots, N$. (P_i^* = number of USD you need, S = price of 1 USD in terms of KRW)

- By assumption 4,

$$\begin{array}{r} W_1 P_1 = W_1 P_1^* S \\ W_2 P_2 = W_2 P_2^* S \\ \vdots \\ W_N P_N = W_N P_N^* S \\ \hline P = P^* S \end{array}$$

Absolute PPP

□ 절대 구매력 평가설

- Finally, the nominal exchange rate (S) is determined as

$$P = P^* S \Leftrightarrow S = \frac{P}{P^*} \quad \text{under assumptions a,b,c, and d} \quad (3.1)$$

⇒ one equation, three variables $\therefore$ two of them should be *exogenous*

⇒ given P and P^* , S is an endogenous variable

⇒ **partial equilibrium approach**

- The prices, P and P^* are exogenously determined and how they are generated is unknown.
- **Limitation:** Absolute PPP does not hold empirically because the assumptions are unrealistic

Relative PPP

□ 상대 구매력 평가설

- It is often found that in some particular countries, there exists a constant, k such that

$$P_t = kP_t^*S_t.$$

where P_t, P_t^* , and S_t are time-varying. For instance, if $P_t > P_t^*S_t$, then $k > 1$. This friction is adjusted by the constant, k

- It does not have a solid theoretical foundation, just holds empirically.

$$\begin{array}{rcl} & P_t & = kP_t^*S_t \\ -\log & P_{t+1} & = kP_{t+1}^*S_{t+1} \\ \hline \log P_{t+1} - \log P_t & = & \log P_{t+1}^* - \log P_t^* + \log S_{t+1} - \log S_t \end{array}$$

$$\underbrace{\log P_{t+1} - \log P_t}_{\pi_{t+1}} = \underbrace{\log P_{t+1}^* - \log P_t^*}_{\pi_{t+1}^*} + \underbrace{\log S_{t+1} - \log S_t}_{s_{t+1}}$$

$$\pi_{t+1} = \pi_{t+1}^* + s_{t+1} \tag{3.2}$$

where s_{t+1} is a depreciation rate

Uncovered Interest Rate Parity

□ 위험 이자율 평형설 (UIP)

- Assumptions

- a No transaction cost (perfect capital mobility)

- b Risk-neutral agents

- Derivation: Suppose that an investor has 1 KRW and two investment options

- (i) Domestic investment (risk-free)

- : payoff after 1 year = $1 \times (1 + i_t)$

- (ii) Foreign investment (risky, because exchange rate is uncertain)

- : payoff after 1 year = $1 \times \frac{1}{S_t} \times (1 + i_t^*) \times S_{t+1}^e$

⇒ Due to the assumption b, options (i) and (ii) are indifferent to the investor, which leads to the UIP equation as

$$S_t = \frac{1 + i_t^*}{1 + i_t} \times S_{t+1}^e \quad (3.3)$$

Uncovered Interest Rate Parity

□ 위험 이자율 평형설 (UIP)

- Meaning of S_{t+1}^e : Take log on both sides of the UIP equation as

$$S_t = \frac{1 + i_t^*}{1 + i_t} \times S_{t+1}^e$$

$$\Rightarrow \log S_t = i_t^* - i_t + \log S_{t+1}^e$$

Similarly, we can have

$$\log S_{t+1} = i_{t+1}^* - i_{t+1} + \log S_{t+2}^e.$$

Uncovered Interest Rate Parity

□ 위험 이자율 평형설 (UIP)

- Now suppose that we are at time t . As the UIP holds at each time point,

$$\begin{aligned}\log S_t &= i_t^* - i_t + \log S_{t+1}^e \\ \log S_{t+1}^e &= i_{t+1}^{*e} - i_{t+1}^e + \log S_{t+2}^e \\ \log S_{t+2}^e &= i_{t+2}^{*e} - i_{t+2}^e + \log S_{t+3}^e\end{aligned}$$

$$+ \quad \quad \quad \vdots \quad \quad \quad \vdots$$

$$\underbrace{\log S_t}_{\text{today's exchange rate}} = \underbrace{(i_t^* - i_t) + (i_{t+1}^{*e} - i_{t+1}^e) + (i_{t+2}^{*e} - i_{t+2}^e) + \dots}_{\text{stream of interest rate differentials across countries}}$$

$\Rightarrow$ even if today's interest rate does not change, as long as future interest rate expectations change, today's exchange rate changes.

$$S_t = \underbrace{\frac{1 + i_t^*}{1 + i_t}}_{\text{today's interest rate}} \times \underbrace{S_{t+1}^e}_{\text{information about future interest rates}}$$

Uncovered Interest Rate Parity

□ UIP의 실증 테스트

- Empirical test of the UIP: The UIP implies

$$\log S_t = i_t^* - i_t + \log S_{t+1}^e$$

and

$$\begin{aligned}\log S_{t+1}^e - \log S_t &= \text{expected depreciation rate} \\ &= i_t - i_t^*\end{aligned}$$

- If the expected exchange rate, S_{t+1}^e is replaced by its realization, S_{t+1} , then

$$s_{t+1} = \underbrace{\frac{S_{t+1} - S_t}{S_t}}_{\text{depreciation rate}} = i_t - i_t^* + \underbrace{\text{error}}_{\because S_{t+1}^e \neq S_{t+1}}$$

Uncovered Interest Rate Parity

□ UIP의 실증 테스트

- Regression analysis: $s_{t+1} = \frac{S_{t+1} - S_t}{S_t} = \beta_1 + \beta_2(i_t - i_t^*) + error$
 - ⇒ If $\beta_1 = 0, \beta_2 = 1$, then UIP holds
 - ⇒ In reality, $\hat{\beta}_1 \neq 0$ and $\hat{\beta}_2 \ll 1$ (around 0, or sometimes negative)
- Why UIP does not hold empirically?
 - ⇒ Perhaps it is because the assumption "risk-neutral agents" is unrealistic. The assumption "no transaction cost" is plausible, too.

Covered Interest Rate Parity

□ 무위험 이자율 평형설 (CIP)

• Assumptions

- No transaction cost (perfect capital mobility)
- No default risk
- Risk-neutral preference is no longer required

• Forward exchange rate

$F_{t,t+1}$: contract on t , delivery on $t + 1 \Rightarrow$ no uncertainty, and no risk

Example a contract to change 1 USD for 1,000 KRW after a year

$$\Rightarrow F_{t,t+1} = 1,000 \text{ KRW}$$

$$\underbrace{1 + i_t}_{\text{payoff of domestic investment with 1KRW}} = \underbrace{\frac{1}{S_t}(1 + i_t^*)F_{t,t+1}}_{\text{payoff of foreign investment}} \quad (3.4)$$

Since there is no expectation term, nothing is random $\Rightarrow$ risk-free

Covered Interest Rate Parity

□ 무위험 이자율 평형설 (CIP)

• CIP equation

$$\underbrace{1 + i_t}_{\text{payoff of domestic investment with 1KRW}} = \underbrace{\frac{1}{S_t}(1 + i_t^*)F_{t,t+1}}_{\text{payoff of foreign investment}} \quad (3.5)$$

$\Rightarrow$

$$\text{CIP : } \underbrace{F_{t,t+1}}_{\text{endogenous}} = \frac{1 + i_t}{1 + i_t^*} \times \underbrace{S_t}_{\text{exogenous}} \quad (3.6)$$

• Difference from UIP

- In UIP S_t is an endogenous variable
- In CIP $F_{t,t+1}$ is endogenous whereas S_t is exogenous

Covered Interest Rate Parity

□ 실증 분석

- Empirical analysis

- does not hold usually empirically, while CIP does.
- If there is a default risk like in a forward contract between developing and developed countries, CIP does not hold.

Forward Premium Puzzle

□ Forward Premium

- Forward premium

– Definition: $\frac{F_{t,t+1} - S_t}{S_t} = \log F_{t,t+1} - \log S_t$

$$(\because \log \frac{F_{t,t+1}}{S_t} = \log \frac{F_{t,t+1} - S_t + S_t}{S_t} = \log(1 + \frac{F_{t,t+1} - S_t}{S_t}) \approx \frac{F_{t,t+1} - S_t}{S_t})$$

According to CIP, $\log F_{t,t+1} - \log S_t = i_t - i_t^*$

Forward Premium Puzzle

□ Forward Premium Puzzle

- From the regression equation $s_{t+1} = \beta_1 + \beta_2 \underbrace{(i_t - i_t^*)}_{\text{forward premium}} + \text{error}$, the coefficients are estimated as $\beta_1 \neq 0$ and $\beta_2 \ll 1$
- The reason why UIP does not hold is not clear (may be because of home bias, regulations, psychological impacts, risk-neutral agent assumption, etc.)
- Suppose UIP and CIP hold in reality:

$$S_t = \frac{1 + i_t^*}{1 + i_t} S_{t+1}^e$$

and

$$S_t = \frac{1 + i_t^*}{1 + i_t} F_{t,t+1}.$$

Then, $S_{t+1}^e = F_{t,t+1}$.

Forward Premium Puzzle

□ Forward Premium Puzzle

- If you want to obtain an expected future exchange rate (i.e., S_{t+1}^e = market expectation), you can just look up the forward exchange rate which can be known from the forward market.
- *However, UIP does not hold in reality.* That is, $S_{t+1}^e \neq F_{t,t+1}$. Therefore, $F_{t,t+1}$ is **not** a good proxy of S_{t+1}^e .

Forward Premium Puzzle

□ Forward Premium Puzzle

[Note] Consider the following statement: the UIP equation,

$$S_t = \frac{1 + i_t^*}{1 + i_t} S_{t+1}^e$$

indicates that i_t rises by 1% in response to the rise of i_t^* by 1%

- This statement is not quite correct. It is because S_t is the only endogenous variable
- Since it is a partial equilibrium, only S_t fluctuates and other variables should be exogenously given. Exogenous variables cannot be affected by other exogenous variables. If they do, another structural equation is needed
- In other words, we do not know how i_t is determined, so it is not sensible to say that i_t goes up just because i_t^* goes up

Fisher Equation

□ 피셔 방정식

• Assumptions

- (a) risk-neutral agents
- (b) no arbitrage condition
- (c) The real return is non-random i.e., the real return is known at present ($r_{t+1}^e = r_t$)

• Derivation

- K_t : quantity of real goods (measured in real variables)
- There are two investment options with K_t
 - (i) sell K_t at price P_t
 - $\Rightarrow P_t K_t$ cash $\Rightarrow$ deposit $\Rightarrow$ 1 year later, $(1 + i_t)P_t K_t$ payoff
 - (ii) keep K_t for a year
 - $\Rightarrow$ at time $t + 1$, K_{t+1} (productivity changes) at P_{t+1} and sell it
 - $\Rightarrow$ 1 year later, $K_{t+1}P_{t+1}$ payoff

Fisher Equation

□ 피셔 방정식

$$\begin{aligned}\therefore \underbrace{P_t K_t (1 + i_t)}_{\text{risk-free}} &= \underbrace{P_{t+1}^e K_{t+1}^e}_{\text{risky}} \\ \Rightarrow \underbrace{1 + i_t}_{\text{nominal returns}} &= \frac{P_{t+1}^e}{P_t} \underbrace{\frac{K_{t+1}^e}{K_t}}_{\text{real returns}}\end{aligned}$$

$\Rightarrow$ take log to have

$$i_t = \pi_{t+1}^e + r_{t+1}^e = \pi_{t+1}^e + r_t.$$

By the assumption that $r_{t+1}^e = r_t$, it follows that

$$i_t = \pi_{t+1}^e + r_t : \text{Fischer Equation} \quad (3.7)$$

Relation among PPP, UIP and Fisher Equation

□ PPP, UIP와 Fisher 방정식

- Assumptions

- (a) real interest rates are the same across countries : $r_t = r_t^*$
- (b) no uncertainty

- Two of the followings imply the other

- (a) Relative PPP: $s_t = \pi_t - \pi_t^*$
- (b) UIP: $s_{t+1} = i_t - i_t^*$
- (c) Fisher Equation: $i_t = r_t + \pi_{t+1}$

Relation among PPP, UIP and Fisher Equation

□ 상대 PPP + 피셔 방정식 $\Rightarrow$ UIP

1. Relative PPP + Fisher equation imply UIP

As the Fischer equation holds for both countries,

$$\begin{aligned} i_t &= r_t + \pi_{t+1} \\ - \quad \underline{i_t^* &= r_t^* + \pi_{t+1}^*}. \end{aligned}$$

This yields that

$$\begin{aligned} i_t - i_t^* &= r_t - r_t^* + \pi_{t+1} - \pi_{t+1}^* \\ &= \pi_{t+1} - \pi_{t+1}^* \quad (\because r_t = r_t^*) \\ &= s_{t+1} \quad (\because PPP) \end{aligned}$$

$\therefore s_{t+1} = i_t - i_t^* : \text{UIP}$

Relation among PPP, UIP and Fisher Equation

□ UIP + 피셔 방정식 $\Rightarrow$ 상대 PPP

2. UIP + Fisher equation imply Relative PPP

by UIP,

$$\begin{aligned} s_{t+1} &= i_t - i_t^* \\ &= r_t + \pi_{t+1} - i_t \quad (\because \text{Fisher Equation}) \\ &= \pi_{t+1} - \pi_{t+1}^* \quad (\because i_t^* = r_t^* + \pi_{t+1}^*) \end{aligned}$$

$$\therefore s_{t+1} = \pi_{t+1} - \pi_{t+1}^* : \text{PPP}$$

Relation among PPP, UIP and Fisher Equation

□ $UIP + PPP \Rightarrow$ 피셔 방정식

3. $UIP + PPP$ imply the Fisher equation: Homework

[참고] 종합적 관계

- 종합적 관계 (표기는 다르지만 충분히 유추할 수 있음)

